

Evaluating Resource Scaling in Containerized Kubernetes Deployments

Abstract:

Abstract: We present an analysis of auto-scaling in cloud computing environments. By examining virtual machines, microservices, and serverless execution latency on Kubernetes cluster configurations, we optimize scaling delays and container lifecycle workflows. We deploy applications across GCP and AWS infrastructure.

Introduction:

In modern systems, this domain has seen significant growth. Traditional methods often struggle with scalability and performance limitations. We introduce our approach, which seeks to resolve these issues by introducing novel methods. We validate our design through extensive simulations and comparative tests. The following sections will detail our implementation design, configurations, and results.